

DAIVIK WADHWANI

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[LinkedIn](#)

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[Leetcode](#)

[Portfolio](#)

Education

Indian Institute of Information Technology Sricity

August 2023 - June 2027

Bachelor of Technology in Computer Science (CGPA of 7.76)

Experience

Research Intern, HCI Lab

Jan 2026 – Present

Indian Institute of Information Technology Sricity

- Co-authored a research paper on one-handed VR accessibility for players with motor disabilities, accepted to ACM ASSETS 2026, the ACM SIGACCESS Conference on Computers and Accessibility.
- Designed and ran a 17-participant user study applying NASA-TLX (Raw TLX 42.0) and System Usability Scale (65.8) instruments; identified a negative learning slope in 12 of 13 long-session participants as quantitative evidence of motor adaptation.

Projects

SplitSmart  | MERN, Socket.io, Gemini Vision API, JWT

2026

AI-powered group expense splitter with real-time cooperative bill claiming and OCR receipt parsing.

- Added **rate limiting** keyed to authenticated user IDs and **Multer** memory-based file handling to prevent API abuse and DoS attacks.
- Integrated **Gemini 2.5-Flash** for receipt OCR via a zero-disk in-memory pipeline; resolved Socket.io race conditions with **OCC and jitter-based retry backoff**
- Optimised **compound MongoDB indexes** across 4 models — scans **6,021 to 15**, DB time **9ms to 1ms**; added **JWT refresh** and rate limiting

Bookish  | MERN, Redux Toolkit, Stripe, Upstash Redis, Jest

2026

Full-stack book marketplace connecting buyers, sellers, and admins with role-based dashboards.

- Integrated **Upstash Redis cache-aside layer** across 11 endpoints — **84.82% latency reduction**, warm hits under **30ms** (from 120-230ms)
- Built **CI/CD pipeline** with GitHub Actions running **48 Jest tests** across 12 suites on every push
- Added **B-Tree indexing**, field projections, and **.lean()** hydration bypass — cut CPU overhead by **35-50%**

PowerPulse  | AWS Lambda, DynamoDB, API Gateway, Cognito, Serverless

2025

Serverless energy analytics system with rule-based and AI-driven insights.

- Architected a fully **serverless backend** on **AWS Lambda, API Gateway, Cognito, and DynamoDB**
- Eliminated idle infrastructure cost and enabled **automatic scaling** without capacity planning
- Devised a caching strategy — response time **1800ms to 200ms (89% faster)**, **API costs down 70%** via **Hugging Face** fallback inference

Technical Skills

Languages: C, C++, C#, Java, JavaScript, HTML, CSS

Frontend: React, Tailwind CSS, Redux Toolkit, TanStack Query

Backend: Node.js, Express.js, REST APIs, Socket.io, JWT, Multer

Cloud & Infra: AWS Lambda, API Gateway, Cognito, Cloudinary, Render, Vercel

Databases: MongoDB, MySQL, DynamoDB, Upstash Redis

Tools: Git, GitHub, Jest, GitHub Actions, Unity, OpenXR, XR Interaction Toolkit, Postman

Achievements

- Secured 3rd place at Global Game Jam 2025 — shipped a complete game under a 48-hour constraint.
- Rated 1580 on LeetCode; solved 250+ problems across LeetCode and Codeforces.
- XR Lead, IOTA Club — leading XR initiatives, mentoring members, and managing lab projects.